

Competency

- To solve the general problems on ratio and proportion, algebraic forms and behavioural problems

Learning Outcomes

- To solve the problems related to ratio and proportion

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To define ratio - To identify antecedent and consequent of the ratio - To recall the types of ratios: compound, duplicate, sub-duplicate, triplicate, sub-triplicate, inverse ratio - To define proportion - To recall the properties of proportion: invertendo, alternaendo, componendo, dividendo and addendo
2.	Understanding (U)	- To find the simple ratio - To solve the algebraic form of ratio and proportion
3.	Application (A)	- To prove the conditional relations on proportion
4.	High Ability (HA)	- To identify the contextual problems based on ratio and proportion and solve them - To solve the higher level conditional problems on proportion

Required Teaching Materials/ Resources

Chart paper with types of ratios and properties of proportions separately with examples

Pre-knowledge: ratio and proportion**Teaching Activities**

1. Ask the weight of two students and ask to find their ratio, similarly collect the prices of copies, pen or bags, umbrellas etc bought in the classroom by the students and discuss on ratios.
2. Under discussion, list the following notes
 - (i) The comparison of quantities of same kind by division. The ratio represents how many times a quantity is greater or smaller than another quantity of same kind
 - (ii) If a and b are two quantities of same kind, the ratio of a and b is $a:b$ or $\frac{a}{b}$
 - (iii) In the ratio $a:b$, a is called antecedent and b called consequent.

3. Discuss on the types of ratio with examples as follows

(i) Compound ratio:

A new ratio obtained by multiplying two or more ratios is called the compound ratio of the given ratios.

For two ratios $a : b$ and $c : d$, the compound ratio $= (a : b) \times (c : d) = \frac{ac}{bd}$

Example: The compound ratio of 2:3 and 9:10 is

(ii) Duplicate and sub-duplicate ratio:

A new ratio obtained by multiplying a ratio by itself is called the duplicate ratio of the given ratio.

Suppose $a : b$ is a ratio then the duplicate ratio $a : b \times a : b = a^2 : b^2$

Example: The duplicate ratio of 4:5 $= 4^2 : 5^2 = 16 : 25$

A new ratio obtained by taking square root of a ratio is called the sub-duplicate ratio of the given ratio.

Suppose $a : b$ is a ratio then the sub-duplicate ratio is $\sqrt{a} : \sqrt{b}$

Example: The sub-duplicate ratio of 9:49 $= \sqrt{9} : \sqrt{49} = 3 : 7$

(iii) Triplicate and sub-triplicate ratio:

A new ratio obtained by multiplying a ratio three times by itself is called the triplicate ratio of the given ratio.

Suppose $a : b$ is a ratio then the triplicate ratio is $a^3 : b^3$

Example: The triplicate ratio of 2:3 $= 2^3 : 3^3 = 8 : 27$

A new ratio obtained by taking cube root of a ratio is called the sub-triplicate ratio of the given ratio.

Suppose $a : b$ is a ratio then the sub-triplicate ratio $\sqrt[3]{a} : \sqrt[3]{b}$

Example: The sub-triplicate ratio of 125 : 64 $= \sqrt[3]{125} : \sqrt[3]{64} = 5 : 4$

(iv) Inverse ratio:

A new ratio obtained by interchanging the antecedent and consequent is the inverse ratio of the given ratio.

Suppose $a : b$ is a ratio then the inverse ratio is $b : a$

Example: The inverse ratio of 4:7 is 7 : 4

4. With proper guidelines, encourage the students to solve the problems related to ratio given in the exercise

5. With proper examples, discuss upon the proportion, proportional and the relation between the means and extremes

6. Discuss upon the properties of proportions

If a, b, c and d are in proportion then there are the following properties

(i) Invertendo

If $\frac{a}{b} = \frac{c}{d}$, then $\frac{b}{a} = \frac{d}{c}$

Proof

Here, $\frac{a}{b} = \frac{c}{d}$, then $1 \div \frac{a}{b} = 1 \div \frac{c}{d}$ or, $1 \times \frac{b}{a} = 1 \times \frac{d}{c}$ or, $\frac{b}{a} = \frac{d}{c}$ Proved

Example: For 2:3 = 4:6, 3:2 = 2×3:2×2 = 6:4

(ii) Alternendo

If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a}{c} = \frac{b}{d}$

Proof

Here, $\frac{a}{b} = \frac{c}{d}$, then multiplying both sides by $\frac{b}{c}$, we get,

$$\frac{a}{b} \times \frac{b}{c} = \frac{c}{d} \times \frac{b}{c} \quad \text{or,} \quad \frac{a}{c} = \frac{b}{d} \quad \text{Proved}$$

Example: For 2:5 = 4:10, 2:4 = 1:2 = 5×1:5×2 = 5:10

(iii) Componendo

$$\text{If } \frac{a}{b} = \frac{c}{d}, \text{ then } \frac{a+b}{b} = \frac{c+d}{d}$$

Proof

Here, $\frac{a}{b} = \frac{c}{d}$, then adding 1 to both sides,

$$\frac{a}{b} + 1 = \frac{c}{d} + 1 \quad \text{or, } \frac{a+b}{b} = \frac{c+d}{d} \quad \text{Proved}$$

Example: For 3:5 = 6:10, (3+5):5 = 8:5 and (6+10):10 = 16:10 = 8:5

Thus, 3:5 = 6:10 implies (3+5):5 = (6+10):10

(iv) Dividendo

$$\text{If } \frac{a}{b} = \frac{c}{d}, \text{ then } \frac{a-b}{b} = \frac{c-d}{d}$$

Proof

Here, $\frac{a}{b} = \frac{c}{d}$, then subtracting 1 from both sides,

$$\frac{a}{b} - 1 = \frac{c}{d} - 1 \quad \text{or, } \frac{a-b}{b} = \frac{c-d}{d} \quad \text{Proved}$$

Example: For 7:4 = 21:12, (7-4):4 = 3:4 and (21-12):12 = 9:12 = 3:4

Thus, 7:4 = 21:12 implies (7-4):4 = (21-12):12

(v) Componendo and Dividendo

$$\text{If } \frac{a}{b} = \frac{c}{d}, \text{ then } \frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Proof

Here, $\frac{a}{b} = \frac{c}{d}$, then by componendo, we have, Also, by dividendo, we have

$$\frac{a+b}{b} = \frac{c+d}{d} \quad \text{..... (i)} \quad \text{or, } \frac{a-b}{b} = \frac{c-d}{d} \quad \text{..... (ii)}$$

Dividing equation (i) by (ii), we get

$$\frac{\frac{a+b}{b}}{\frac{a-b}{b}} = \frac{\frac{c+d}{d}}{\frac{c-d}{d}} \quad \text{or, } \frac{a+b}{a-b} = \frac{c+d}{c-d} \quad \text{proved}$$

Example: For 3:2 = 9:6, (3+2):(3-2) = 5:1 and (9+6):(9-6) = 15:3 = 5:1

Thus, 3:2 = 9:6 implies (3+2):(3-2) = (9+6):(9-6)

(vi) Addendo

$$\text{If } \frac{a}{b} = \frac{c}{d}, \text{ then } \frac{a}{b} = \frac{c}{d} = \frac{a+c}{b+d}$$

Proof

Here, $\frac{a}{b} = \frac{c}{d}$, then by alternendo, we have, $\frac{a}{c} = \frac{b}{d}$

and by componendo, we have $\frac{a+c}{c} = \frac{b+d}{d}$

Again, by alternaendo, we have, $\frac{a+c}{b+d} = \frac{c}{d} \quad \therefore \frac{a}{b} = \frac{c}{d} = \frac{a+c}{b+d}$ proved.

Example: For 5:7 = 10:14, (5+10):(7+14) = 15:21 = 5:7

Thus, $5:7 = 10:14 = (5 + 10):(7 + 14)$

7. Prove the conditional identities on proportion with discussion and give some similar problems to prove in the class.

Solution of selected problems from Vedanta Excel in Mathematics

1. If $a^2 - 2ab + b^2 : a^2 + 2ab + b^2 = 1 : 4$, find $a : b$.

Solution:

$$\frac{a^2 - 2ab + b^2}{a^2 + 2ab + b^2} = \frac{1}{4}$$

or, $\frac{(a-b)^2}{(a+b)^2} = \frac{1}{4}$

or, $\left(\frac{a-b}{a+b}\right)^2 = \left(\pm\frac{1}{2}\right)^2$

Taking (+) ve sign

$$\frac{a-b}{a+b} = \frac{1}{2}$$

or, $2a - 2b = a + b$

or, $a = 3b$

or, $\frac{a}{b} = \frac{3}{1}$

$$\therefore a : b = 3 : 1$$

Taking (-) ve sign

$$\frac{a-b}{a+b} = \frac{-1}{2}$$

or, $2a - 2b = -a - b$

or, $3a = b$

or, $\frac{a}{b} = \frac{1}{3}$

$$\therefore a : b = 1 : 3$$

2. If $a : b = 3 : 2$, find the value of

i) $b - \frac{2a}{3}$

ii) $\frac{a^2 - ab}{ab + b^2}$

Solution:

$$a : b = 3 : 2$$

$$\text{Let } a = 3x \text{ and } b = 2x$$

i) $b - \frac{2a}{3} = 2x - \frac{2 \times 3x}{3} = 0$

ii) $\frac{a^2 - ab}{ab + b^2} = \frac{(3x)^2 - 3x \times 2x}{3x \cdot 2x + (2x)^2}$

$$= \frac{9x^2 - 6x^2}{6x^2 + 4x^2}$$

$$= \frac{3x^2}{10x^2}$$

$$= \frac{3}{10} = 3 : 10$$

3. If $(x + 4) : (3x + 1)$ is the duplicate ratio of $3 : 4$, find the value of x .

Solution:

$$\text{Duplicate of } 3 : 4 = 3^2 : 4^2 = 9 : 16$$

By question,

$$\frac{x+4}{3x+1} = \frac{9}{16}$$

or, $16x + 64 = 27x + 9 \quad \therefore x = 5$

4. If $(3a - 7) : (4a + 3)$ is the sub-triplicate ratio of $8 : 27$, find the value of a .

Solution:

Here, sub - triplicate ratio of $8 : 27 = 2 : 3$.

According to question,

Sub - triplicate ratio of $8 : 27 = (3a - 7) : (4a + 3)$

$$\text{or, } \frac{2}{3} = \frac{3a - 7}{4a + 3}$$

$$\text{or, } 9a - 21 = 8a + 6 \quad \therefore a = 27$$

5. The ratio of two number is $2 : 3$ and their lcm is 30, find the number.

Solution:

Let the required number be $2x$ and $3x$.

According to the question;

LCM of $2x$ and $3x = 30$

$$\text{or, } x \times 2 \times 3 = 30 \quad \therefore x = 5$$

$$\text{Hence, the first number} = 2x = 2 \times 5 = 10$$

$$\text{and the second number} = 3x = 3 \times 5 = 15$$

6. In 50 l of milk, the ratio of pure milk and water is $2 : 3$. How much pure milk should be added to the mixture so that the pure milk and water will be $5 : 6$ ratio ?

Solution:

Let the quantity of pure milk in 50 l of milk be $2x$ l and the quantity of water be $3x$ l.

$$\text{Now, } 2x + 3x = 50 \text{ l} \quad \therefore x = 10 \text{ l}$$

$$\therefore \text{Quantity of pure milk} = 2x = 2 \times 10 = 20 \text{ l}$$

$$\text{and Quantity of water} = 3x = 3 \times 10 = 30 \text{ l}$$

Again,

Let the quantity of pure milk to be added in the mixture be y l.

Then, according to question:

$$\frac{20 + y}{30} = \frac{5}{6}$$

$$\text{or, } y = 5$$

Hence, the quantity of milk to be added is 5 l.

7. If $(a - b)$, b and $(a + b)$ are in continuous proportion, show that $\frac{a^2}{b^2} = 2$.

Solution:

Here, $(a - b)$, b and $(a + b)$ are in continuous.

$$\text{so, } b^2 = (a - b)(a + b)$$

$$\text{or, } b^2 = a^2 - b^2$$

$$\text{or, } 2b^2 = a^2$$

$$\text{or, } 2 = \frac{a^2}{b^2} \quad \therefore \frac{a^2}{b^2} = 2 \text{ proved}$$

8. What number should be added to each term 7, 10, 16 and 22. So that they will be in proportion ?

Solution:

Let the number to be added be x .

$$\text{Then, } \frac{7 + x}{10 + x} = \frac{16 + x}{22 + x}$$

$$\text{or, } 154 + 7x + 22x + x^2 = 160 + 10x + 16x + x^2$$

$$\text{or, } 154 + 29x = 26x + 160$$

$$\text{or, } 3x = 6 \quad \therefore x = 2$$

Hence, the required number to be added is 2.

9. If $\frac{a}{b} = \frac{a}{b}$, prove that:

a) $\frac{a^2 + ab + b^2}{b^2 + bc + c^2} = \frac{a}{c}$

b) $\frac{a^3 + b^3}{b^3 + c^3} = \frac{a(a-b)}{c(b-c)}$

c) $\frac{a+b+c}{a-b+c} = \frac{(a+b+c)^2}{a^2+b^2+c^2}$

Solution:

Let $\frac{a}{b} = \frac{b}{c} = k$ then $\frac{b}{c} = k$ i.e, $b = ck$

and $\frac{a}{b} = k$ i.e, $a = bk = ck.k = ck^2$

Now

a) L.H.S = $\frac{a^2 + ab + b^2}{b^2 + bc + c^2} = \frac{(ck^2)^2 + ck^2.ck + (ck)^2}{(ck)^2 + ck.c + c^2} = \frac{c^2k^4 + c^2k^3 + c^2k^2}{c^2k^2 + c^2k + c^2} = \frac{c^2k^2(k^2 + k + 1)}{c^2(k^2 + k + 1)} = k^2$

R.H.S = $\frac{a}{c} = \frac{ck^2}{c} = k^2$

Hence, L.H.S = R.H.S Proved

b) L.H.S = $\frac{a^3 + b^3}{b^3 + c^3} = \frac{(ck^2)^3 + (ck)^3}{(ck)^3 + c^3} = \frac{c^3k^6 + c^3k^3}{c^3k^3 + c^3} = \frac{c^3k^3(k^3 + 1)}{c^3(k^3 + 1)} = k^3$

R.H.S = $\frac{a(a-b)}{c(b-c)} = \frac{ck^2(ck^2 - ck)}{c(ck - c)} = \frac{k^2.ck(k-1)}{c(k-c)} = k^3$

Hence, L.H.S = R.H.S Proved

c) L.H.S = $\frac{a+b+c}{a-b+c} = \frac{ck^2 + ck + c}{ck^2 - ck + c} = \frac{c(k^2 + k + 1)}{c(k^2 - k + 1)} = \frac{(k^2 + k + 1)}{(k^2 - k + 1)}$

R.H.S = $\frac{(a+b+c)^2}{a^2 + b^2 + c^2}$
 $= \frac{(ck^2 + ck + c)^2}{(ck^2)^2 + (ck)^2 + c^2}$
 $= \frac{c^2(k^2 + k + 1)^2}{c^2k^4 + c^2k^2 + c^2}$
 $= \frac{c^2(k^2 + k + 1)^2}{c^2(k^4 + k^2 + 1)}$
 $= \frac{(k^2 + k + 1)^2}{(k^2 + 1)^2 - 2k^2 + k^2} = \frac{(k^2 + k + 1)^2}{(k^2 + 1)^2 - k^2} = \frac{(k^2 + k + 1)^2}{(k^2 + k + 1)(k^2 - k + 1)} = \frac{(k^2 + k + 1)}{(k^2 - k + 1)}$

10. If x, y, z are in continuous proportion. Prove that $x^2y^2z^2\left(\frac{1}{x^3} + \frac{1}{y^3} + \frac{1}{z^3}\right) = x^3 + y^3 + z^3$.

Solution:

Let $\frac{x}{y} = \frac{y}{z} = k$ then $\frac{y}{z} = k$ i.e $y = zk$,

$\frac{x}{y} = k$ i.e, $x = yk = zk.k = zk^2$

Now, L.H.S = $x^2y^2z^2\left(\frac{1}{x^3} + \frac{1}{y^3} + \frac{1}{z^3}\right)$

$= (3k^2)^2 (3k)^2 z^2 \left(\frac{1}{(zk^2)^3} + \frac{1}{(zk)^3} + \frac{1}{z^3}\right)$

$= z^2k^4 . z^2k^2 . z^2 \left(\frac{1}{z^3k^6} + \frac{1}{z^3k^3} + \frac{1}{z^3}\right) = z^6k^6 \left(\frac{1 + k^3 + k^6}{z^3k^6}\right) = z^3(1 + k^3 + k^6)$

R.H.S = $x^3 + y^3 + z^3 = (3k^2)^3 + (3k)^3 + z^3 = z^3k^6 + z^3k^3 + z^3 = z^3(k^6 + k^3 + 1) = z^3(1 + k^3 + k^6)$

Hence, L.H.S = R.H.S Proved

11. If $\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$, prove that: $\frac{a^2 - ab}{b^2} = \frac{b^2 - bc}{c^2} = \frac{c^2 - cd}{d^2}$

Solution:

Let $\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = k$ then $\frac{c}{d} = k$ i.e. $c = dk$

$\frac{b}{c} = k$ i.e. $b = ck = dk.k = dk^2$

and $\frac{a}{b} = k$ i.e. $a = bk = dk^2.k = dk^3$

Now,

$$\begin{aligned} \text{L.H.S} &= \frac{a^2 - ab}{b^2} = \frac{(dk^3)^2 - dk^3 \cdot dk^2}{(dk^2)^2} \\ &= \frac{d^2k^6 - d^2k^5}{d^2k^4} = \frac{d^2k^5(k-1)}{d^2k^4} = k(k-1) \end{aligned}$$

$$\text{Middle Term (M.T.)} = \frac{b^2 - bc}{c^2} = \frac{(dk^2)^2 - dk^2 \cdot dk}{(dk)^2} = \frac{d^2k^4 - d^2k^3}{d^2k^2} = \frac{d^2k^3(k-1)}{d^2k^2} = k(k-1)$$

$$\text{R.H.S} = \frac{c^2 - cd}{d^2} = \frac{(dk)^2 - dk \cdot d}{d^2} = \frac{d^2k(k-1)}{d^2} = k(k-1)$$

Hence, L.H.S. = M.T. = R.H.S. Proved

12. If $a : b :: b : c :: c : d$, prove that : $(a - c)^2 + (b - c)^2 + (b - d)^2 = (a - d)^2$

Solution:

Here, $\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$

$\therefore ac = b^2, bd = c^2$ and $ad = bc$

$$\begin{aligned} \text{LHS} &= (a - c)^2 + (b - c)^2 + (b - d)^2 \\ &= a^2 - 2ac + c^2 + b^2 - 2bc + c^2 + b^2 - 2bd + d^2 \\ &= a^2 - 2b^2 + c^2 + 2b^2 - 2ad + c^2 - 2c^2 + d^2 = a^2 - 2ad + d^2 = (a - d)^2 \end{aligned}$$

13. If $x : a :: y : b :: z : c$, prove that: $\frac{ax - by}{(a + b)(x - y)} + \frac{by - cz}{(b + c)(y - z)} + \frac{cz - ax}{(c + a)(z - x)} = 3$

Solution:

Let, $\frac{x}{a} = \frac{y}{b} = \frac{z}{c} = k$ $\therefore x = ak, y = bk$ and $z = ck$

$$\begin{aligned} \text{LHS} &= \frac{ax - by}{(a + b)(x - y)} + \frac{by - cz}{(b + c)(y - z)} + \frac{cz - ax}{(c + a)(z - x)} \\ &= \frac{a.ak - b.bk}{(a + b)(ak - bk)} + \frac{b.bk - c.ck}{(b + c)(bk - ck)} + \frac{c.ck - a.ak}{(c + a)(ck - ak)} \\ &= \frac{k(a^2 - b^2)}{k(a + b)(a - b)} + \frac{k(b^2 - c^2)}{k(b + c)(b - c)} + \frac{k(c^2 - a^2)}{k(c + a)(c - a)} \\ &= 1 + 1 + 1 = 3 = \text{R.H.S. Proved} \end{aligned}$$

14. If $(a^2 + b^2)(x^2 + y^2) = (ax + by)^2$, show that: $\frac{x}{a} = \frac{y}{b}$

Solution:

Here, $(a^2 + b^2)(x^2 + y^2) = (ax + by)^2$

$$\text{or, } a^2x^2 + a^2y^2 + b^2x^2 + b^2y^2 = a^2x^2 + 2abxy + b^2y^2$$

$$\text{or, } a^2y^2 - 2abxy + b^2x^2 = 0$$

$$\text{or, } (ay - bx)^2 = 0$$

$$\text{or, } ay - bx = 0$$

$$\text{or, } ay = bx$$

$$\text{or, } \frac{y}{b} = \frac{x}{a}$$

Hence, $\frac{x}{a} = \frac{y}{b}$ Proved

15. If $\frac{ay - bx}{p} = \frac{cx - az}{q} = \frac{bz - cy}{r}$, Prove that $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$

Solution:

$$\frac{ay - bx}{p} = \frac{cx - az}{q} = \frac{bz - cy}{r}$$

or, $\frac{c(ay - bx)}{cp} = \frac{b(cx - az)}{bq} = \frac{a(bz - cy)}{ar}$

We know,

$$\text{Each ratio} = \frac{\text{Sum of antecedents}}{\text{Sum of consequents}}$$

$$\text{or, } \frac{c(ay - bx)}{cp} = \frac{b(cx - az)}{bq} = \frac{a(bz - cy)}{ap} = \frac{c(ay - bx) + b(cx - az) + a(bz - cy)}{cp + bq + ap}$$

$$\text{or, } \frac{c(ay - bx)}{cp} = \frac{b(cx - az)}{bq} = \frac{a(bz - cy)}{ap} = 0$$

$$\text{or, } \frac{c(ay - bx)}{cp} = 0, \frac{b(cx - az)}{bq} = 0 \text{ and } \frac{a(bz - cy)}{ap} = 0$$

$$\text{or, } ay - bx = 0, cx - az = 0 \text{ and } bz - cy = 0$$

$$\text{or, } ay = bx, cx = az \text{ and } bz = cy$$

$$\text{or, } \frac{y}{b} = \frac{x}{a}, \frac{x}{a} = \frac{z}{c} \text{ and } \frac{z}{c} = \frac{y}{b}$$

$$\text{or, } \frac{x}{a} = \frac{y}{b} = \frac{z}{c}$$

Hence proved

Extra Question

1. If $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, prove that: $\frac{x^3 + y^3 + z^3}{a^3 + b^3 + c^3} = \frac{xyz}{abc}$
2. If $\frac{a}{b} = \frac{b}{c}$, prove that: $\frac{a + b}{b + c} = \frac{a^2(b - c)}{b^2(a - b)}$
3. If a, b, c are in continuous proportion, prove that $(ab + bc + ca)^3 = abc(a + b + c)^3$.
4. If $a : b :: b : c :: c : d$, prove that $\frac{a^3 + b^3 + c^3}{b^3 + c^3 + d^3} = \frac{a^3 + b^3 + abc}{b^3 + c^3 + bcd}$
5. If a, b, c, d and e are in continuous proportion, prove that $a : e = a^4 : b^4$